PES University - Dept. of CSE

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Name: Jayant Mahishi **SRN:** PES1UG24AM122

Scenario: Problem Statement #16 - Remote Patient Vitals Alert & Monitoring App (Healthcare & Telemedicine)

Source: Functional requirements FR-001 to FR-005 and NFR-001/002 from Lab 1

1. Epics

Epic	Name	Description	Lab 1 Req.	Stories	Points
EPIC-1	Telemetry Ingestion & Anomaly Detection	Continuously ingest SpO2, Heart Rate and BP from the patient's wearable, validate readings and detect threshold breaches within 2 seconds on a scalable, secure gateway.	FR-001, NFR-001	4	24
EPIC-2	Emergency Alerting & Caregiver Escalation	Raise emergency alerts for critical breaches, notify the on-call caregiver within 3 seconds and escalate through the caregiver matrix reliably.	FR-002, FR-003, NFR-002	4	21
EPIC-3	Caregiver Response & Acknowledgement	Let the on-call caregiver view alert details and acknowledge an alert in one action, stopping escalation and keeping an audit trail.	FR-004	3	8
EPIC-4	Monitoring Dashboard, Configuration & Privacy	Give patients and caregivers live visibility of vitals and alerts, let caregivers configure patient thresholds, and protect health data with access control.	FR-005, FR-001, NFR-002	5	23
	Total			**16**	**76**

2. User Stories, Priority and Story Points (Fibonacci)

ID	Story (As a / I want / So that)	Priority	Pts	Sprint
1.1 EPIC-1	**Stream vitals from wearable** As a remote patient, I want my wearable to continuously send SpO2, heart rate and BP to the system, so that my condition is monitored without hospital visits.	High	5	1
1.2 EPIC-1	**Validate incoming telemetry** As a on-call caregiver, I want corrupted or out-of-range sensor readings to be filtered out, so that false alarms are reduced and I can trust the alerts.	High	3	1
1.3 EPIC-1	**Detect threshold breach within 2 s** As a on-call caregiver, I want critical readings (e.g. HR > 140 BPM) flagged within 2 seconds, so that I can respond before the patient deteriorates.	High	8	1
1.4 EPIC-1	**Secure, scalable ingestion gateway** As a on-call caregiver, I want the gateway to handle 500 concurrent encrypted streams at 99.99% uptime, so that every post-operative patient is monitored reliably and securely.	High	8	2
2.1 EPIC-2	**Create emergency alert record** As a on-call caregiver, I want an emergency alert created with patient, metric, value and time, so that I have the full context of the emergency.	High	3	1
2.2 EPIC-2	**Notify primary caregiver (push + SMS)** As a on-call caregiver, I want to receive a push and SMS within 3 seconds of a critical breach, so that I am informed immediately even if the app is closed.	High	5	1
2.3 EPIC-2	**Escalate unacknowledged alerts** As a remote patient, I want unacknowledged alerts escalated to the next caregiver after 60 seconds, so that help reaches me even if the first caregiver is unreachable.	High	8	2
2.4 EPIC-2	**Retry and voice-call fallback** As a remote patient, I want failed notifications retried and a voice call used as fallback, so that no emergency alert is ever lost.	Medium	5	2
3.1 EPIC-3	**Acknowledge alert in one action** As a on-call caregiver, I want to acknowledge an alert with a single tap, so that escalation stops and others know I am responding.	High	3	1
3.2 EPIC-3	**Audit trail of acknowledgements** As a on-call caregiver, I want my caregiver ID and acknowledgement time recorded, so that there is an accountable clinical and legal record.	Medium	2	1

ID	Story (As a / I want / So that)	Priority	Pts	Sprint
3.3 EPIC-3	**View alert details with current vitals** As a on-call caregiver, I want to see the patient's current vitals when I open an alert, so that I can judge the severity and act quickly.	Medium	3	1
4.1 EPIC-4	**Patient live vitals view** As a remote patient, I want to see my own live vitals and alert status, so that I feel reassured and informed about my recovery.	Medium	5	2
4.2 EPIC-4	**Caregiver multi-patient dashboard** As a on-call caregiver, I want a dashboard of live vitals for my assigned patients, refreshed every 5 s, so that I keep situational awareness across patients.	Medium	5	2
4.3 EPIC-4	**Configure patient thresholds** As a on-call caregiver, I want to set clinical thresholds per patient, so that alerts match each patient's surgery and condition.	High	5	1
4.4 EPIC-4	**View alert history** As a on-call caregiver, I want to browse a patient's past alerts, so that I can spot trends and review responses.	Low	3	2
4.5 EPIC-4	**Encrypt data and enforce access control** As a remote patient, I want my health data encrypted and visible only to authorised caregivers, so that my privacy is protected.	Medium	5	2

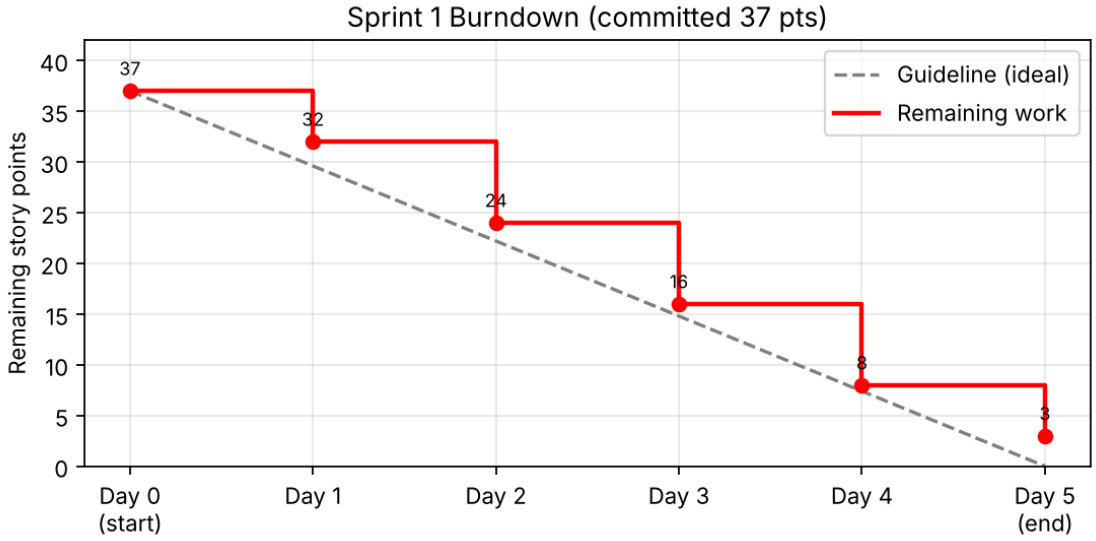
Prioritisation rule: High = directly affects patient safety or the core detect-alert-respond chain; Medium = improves reliability, visibility or privacy; Low = analysis/convenience. **Estimation:** Planning Poker on the Fibonacci scale (1, 2, 3, 5, 8); items of 8 are the most complex (real-time detection, escalation logic, scalable gateway).

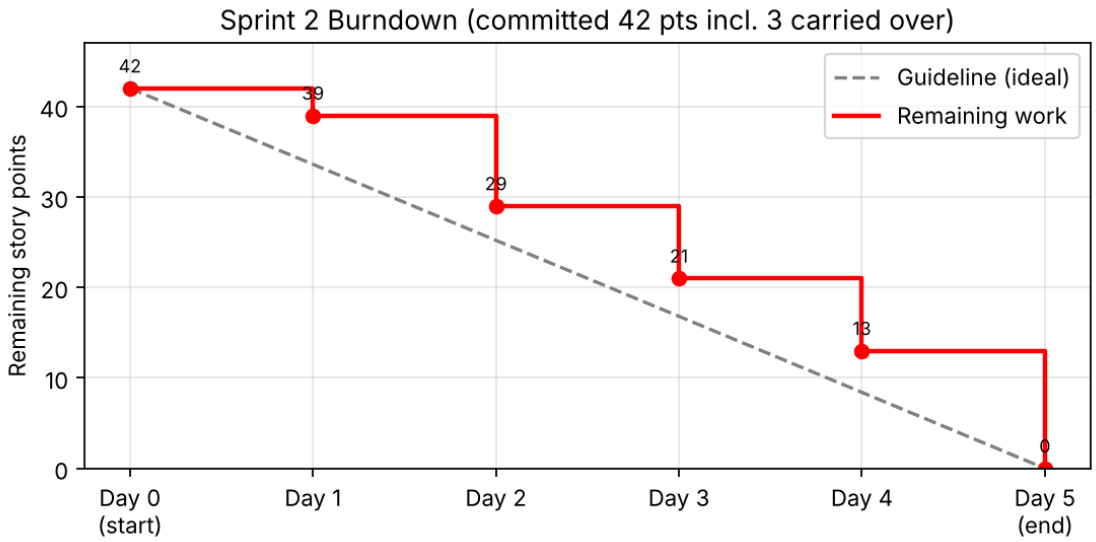
3. Sprint Plan (1 week each)

Sprint	Goal	Stories	Points
Sprint 1	An end-to-end flow: a critical reading is detected, the caregiver is notified and can acknowledge it.	1.1, 1.2, 1.3, 2.1, 2.2, 3.1, 3.2, 3.3, 4.3	37
Sprint 2	Make the system reliable and complete: scalable gateway, escalation, fallbacks, dashboards and privacy.	1.4, 2.3, 2.4, 4.1, 4.2, 4.4, 4.5 (+ 3.3 carried over from Sprint 1 in the simulation)	39 planned / 42 actual

4. Burndown Charts

Note: the charts below are drawn from the planned simulation. Replace or accompany them with the screenshots from Jira (Reports > Burndown Chart) for submission.





5. Reflection Questions

1. Did your estimations reflect the actual effort?
Mostly yes. The 8-point stories (threshold detection, escalation, gateway) were the largest and took the most time, and the 2-3 point stories were completed quickly, so the relative sizing held. The exception was story 3.3 (view alert details with current vitals): it was estimated at 3 but depended on the dashboard data feed, so it was not finished in Sprint 1 and was carried over. Dependencies should be counted as risk when estimating.

2. Was your backlog well-prioritized?
Largely. Safety-critical items (detect, alert, acknowledge, thresholds) were ranked High and scheduled in Sprint 1, giving a working end-to-end flow early. One improvement: the secure gateway (1.4) is High priority but was placed in Sprint 2 because of its size; splitting it into smaller stories (authentication, TLS, load handling) would allow part of it in Sprint 1.

3. How did your simulated sprint align with your plan?
Sprint 1 completed 34 of 37 planned points (about 92%); one 3-point story spilled over. Sprint 2 had 42 points (39 planned plus the carried-over story) and was completed in full. Overall all 76 points were delivered across two sprints, with Sprint 1 slightly over-committed and Sprint 2 slightly heavy at the start of the sprint.

4. What insights did the burndown chart give about your team's capacity?
The Sprint 1 line stayed close to, then slightly above, the ideal guideline and ended with 3 points remaining, which suggests the team's realistic capacity is about 34-35 points per week, so commitments should be about 32-35 points rather than 37. Sprint 2 burned down unevenly: little progress early and a steep drop at the end, which signals work was not broken into small enough pieces and finished late. Smaller stories and finishing items daily would give a smoother burndown. Note that in a classroom simulation where stories are moved to Done within minutes, the real Jira chart can look like a sharp cliff rather than a smooth line.

6. Jira Screenshots (paste from your workspace)

(a) Backlog with Epics and User Stories (b) Story point assignments (c) Active Sprint board (d) Burndown chart (Reports > Burndown Chart)

